

Solutions for Problem Set 3

Mathematics for Social Scientists

EXERCISE 1. Compute the following derivatives (*J. Gill, 2006, Exercise 5.5*):

- i. $\frac{d}{dx} 3x^{1/3}$
- ii. $\frac{d}{dy} y^3 + 3y^2 - 12$
- iii. $\frac{d}{dx} (x^2 + 1)(x^3 - 1)$
- iv. $\frac{d}{dy} (y - y^{-1})(y - y^{-2})$
- v. $\frac{d}{dy} e^{y^2-3y+2}$
- vi. $\frac{d}{dx} \left(\frac{1}{100} x^{25} - \frac{1}{10} x^{0.25} \right)$
- vii. $\frac{d}{dy} (y^3 - 7) \left(1 + \frac{1}{y^2} \right)$
- viii. $\frac{d}{dx} \left(\frac{4x-12x^2}{x^3-4x^2} \right)$
- ix. $\frac{d}{dx} \log(2\pi x^2)$

- i. $x^{-\frac{2}{3}}$
- ii. $3y^2 + 6y$
- iii. $5x^4 + 3x^2 - 2x$
- iv. $2y + y^{-2} - 3y^{-4}$
- v. $e^{y^2-3y+2} \cdot (2y - 3)$
- vi. $\frac{1}{4} x^{24} - \frac{1}{40} x^{-0.75}$
- vii. $3y^2 + 1 + 14 \frac{1}{y^3}$
- viii. $\frac{-12x^2+16x+16}{x^2(x-4)^2}$
- ix. $\frac{2}{x}$

EXERCISE 2. Differentiate the following functions with respect to x .

- (i) $f(x) = 7x^3 - 2x^2 - 5x + 1$
- (ii) $f(x) = 0.7x^{-4} + 1.3 - 3.1x^3$

$$\text{(iii)} \quad f(x) = \frac{3x^2 + 1}{2x}$$

$$\text{(iv)} \quad f(x) = \sqrt{4x + 9}$$

$$\text{(v)} \quad f(x) = \frac{x^{1/3} - 2}{(x^5 - 2)^3}$$

$$\text{(vi)} \quad f(x) = \ln\left(\frac{x^2}{x^4 + 1}\right)$$

$$\text{(vii)} \quad f(x) = e^{x^3+x}$$

$$\text{(viii)} \quad f(x) = \frac{1}{e^x + e^{-x}}$$

$$\text{(i)} \quad f'(x) = 21x^2 - 4x - 5$$

$$\text{(ii)} \quad f'(x) = -2.8x^{-5} - 9.3x^2$$

$$\text{(iii)} \quad f'(x) = \frac{3}{2} - \frac{1}{2x^2}$$

$$\text{(iv)} \quad f'(x) = \frac{2}{\sqrt{4x + 9}}$$

$$\text{(v)} \quad f'(x) = \frac{1}{3}x^{-2/3}(x^5 - 2)^{-3} - 15x^4(x^{1/3} - 2)(x^5 - 2)^{-4}$$

$$\text{(vi)} \quad f'(x) = \frac{2}{x} - \frac{4x^3}{x^4 + 1}$$

$$\text{(vii)} \quad f'(x) = (3x^2 + 1)e^{x^3+x}$$

$$\text{(viii)} \quad f'(x) = -\frac{e^x - e^{-x}}{(e^x + e^{-x})^2}$$

EXERCISE 3. We have seen in class that the definition of the derivative is:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Using this definition, compute the derivatives of the following functions.

$$\text{i.} \quad f(x) = 2x^2 - 8x + 15$$

$$\text{ii.} \quad g(x) = \sqrt{3x - 5}$$

$$\text{iii.} \quad p(x) = \frac{x+1}{x}$$

Hint: For (ii), you might want to use

$$\frac{\sqrt{a}-\sqrt{b}}{h} \cdot \frac{\sqrt{a}+\sqrt{b}}{\sqrt{a}+\sqrt{b}} = \frac{a-b}{h \cdot (\sqrt{a}+\sqrt{b})}$$

i. $4x - 8$

ii. $\frac{3}{2\sqrt{3x-5}}$

iii. $\frac{-1}{x^2}$

EXERCISE 4. Using the L'Hospital's Rule, find the following limits. Explain why you can use the L'Hospital's Rule. (*J. Gill, 2006, Exercise 5.8*)

i. $\lim_{x \rightarrow 1} \frac{\sin(1-x)}{x^2-1}$

ii. $\lim_{x \rightarrow \infty} \frac{\log(x)}{x}$

iii. $\lim_{x \rightarrow \infty} \frac{3^x}{x^3}$

iv. $\lim_{y \rightarrow 0} \frac{e^y \sin(y)}{\log(1-y)}$

v. $\lim_{y \rightarrow 0} \frac{(\sin(y))^2 - \sin(y^2)}{y}$

vi. $\lim_{y \rightarrow \infty} \frac{3e^y}{y^3}$

vii. $\lim_{x \rightarrow \infty} \frac{x \log(x)}{x + \log(x)}$

i. $\frac{-1}{2}$

ii. 0

iii. ∞

iv. -1

v. 0

vi. ∞

vii. ∞

EXERCISE 5. Differentiate the following functions $f : \mathbb{R} \rightarrow \mathbb{R}$.

(i) $f_1(x) = 5$

(ii) $f_2(x) = x^4 + x^2$

(iii) $f_3(x) = \sin(x) \cos(x)$

(iv) $f_4(x) = x\sqrt{x}$

(v) $f_5(x) = e^{2x}$

(vi) $f_6(x) = \sin(x^2 - 1)$

(i) 0

(ii) $4x^3 + 2x$

(iii) $\cos(x)^2 - \sin(x)^2$

(iv) $\frac{3}{2}\sqrt{x}$

(v) $2e^{2x}$

(vi) $2x \cos(x^2 - 1)$

EXERCISE 6. Differentiate the following functions on the maximal possible domain of definition.

(i) $g_1(x) = \frac{3x-5}{2x+8}$

(ii) $g_2(x) = \frac{\sin(x)}{\cos(x)}$

(iii) $g_3(x) = \ln(x^3 + 3)$

(iv) $g_4(x) = x^3(\ln(x))^2$

(i) $\frac{17}{2(x+4)^2}$

(ii) $\frac{1}{\cos x^2}$. Note the trigonometric identity $1 = \sin(x)^2 + \cos(x)^2$

(iii) $\frac{3x^2}{x^3+3}$

(iv) $x^2 \ln(x)(3 \ln(x) + 2)$

EXERCISE 7. Calculate the following higher order derivatives:

- i. $\frac{d^3 e^{2x}}{dx^3}$
- ii. $\frac{d^4 \sin(x)}{dx^4}$
- iii. $\frac{d^{67} \sin(x)}{dx^{67}}$
- iv. $\frac{d^{2022} \cos(x)}{dx^{2022}}$

- i. $8e^{2x}$
- ii. $\sin(x)$
- iii. $-\cos(x)$
- iv. $-\cos(x)$

EXERCISE 8. Determine the partial derivatives $\partial_x f$, $\partial_y f$, ∂_{xx} , ∂_{xy} and ∂_{yy} of the following functions (always considered on the maximal possible domain of definition):

- i. $f(x, y) = x^2 y$
- ii. $f(x, y) = xy^2$
- iii. $f(x, y) = e^{xy^3}$
- iv. $f(x, y) = 4 \cdot \frac{x}{y^5}$
- v. $f(x, y) = \cos(xy)$
- vi. $f(x, y) = (2x - y)^2 + \log(xy)$
- vii. $f(x, y) = xy^2(\sin(x) + \sin(y))$

- i.
 - $\partial_x f = 2xy$
 - $\partial_y f = x^2$
 - $\partial_{xx} f = 2y$
 - $\partial_{xy} f = 2x$
 - $\partial_{yy} f = 0$
- ii.
 - $\partial_x f = y^2$

- $\partial_y f = 2xy$
- $\partial_{xx} f = 0$
- $\partial_{xy} f = 2y$
- $\partial_{yy} f = 2x$
- iii. • $\partial_x f = y^3 e^{xy^3}$
- $\partial_y f = 3xy^2 e^{xy^3}$
- $\partial_{xx} f = y^6 e^{xy^3}$
- $\partial_{xy} f = e^{xy^3} (3y^2 + 3xy^5)$
- $\partial_{yy} f = e^{xy^3} (6xy + 9x^2 y^4)$
- iv. • $\partial_x f = \frac{4}{y^5}$
- $\partial_y f = -\frac{20x}{y^6}$
- $\partial_{xx} f = 0$
- $\partial_{xy} f = -\frac{20}{y^6}$
- $\partial_{yy} f = \frac{120x}{y^7}$
- v. • $\partial_x f = -y \sin(xy)$
- $\partial_y f = -x \sin(xy)$
- $\partial_{xx} f = -y^2 \cos(xy)$
- $\partial_{xy} f = -xy \cos(xy) - \sin(xy)$
- $\partial_{yy} f = -x^2 \cos(xy)$
- vi. • $\partial_x f = 4(2x - y) + \frac{1}{x}$
- $\partial_y f = -2(2x - y) + \frac{1}{y}$
- $\partial_{xx} f = 8 - \frac{1}{x^2}$
- $\partial_{xy} f = -4$
- $\partial_{yy} f = 2 - \frac{1}{y^2}$
- vii. • $\partial_x f = y^2(\sin x + \sin y) + xy^2 \cos x$

- $\partial_y f = 2xy(\sin x + \sin y) + xy^2 \cos y$
- $\partial_{xx} f = 2y^2 \cos x - xy^2 \sin x$
- $\partial_{xy} f = 2y(\sin x + \sin y) + y^2 \cos y + 2xy \cos x$
- $\partial_{yy} f = 2x(\sin x + \sin y) + 4xy \cos y - xy^2 \sin y$

EXERCISE 9. Find all first and second (mixed) partial derivatives of the following functions.

(i) $f(x, y) = \ln x \cdot y^2$

(ii) $f(x, y) = \sqrt{2x - y}$

(iii) $f(x, y) = (x + 4y)(e^{-2x} + e^{-3y})$

(i)

$$\partial_x = \frac{y^2}{x}$$

$$\partial_y = 2y \ln x$$

$$\partial_{xx} = -\frac{y^2}{x^2}$$

$$\partial_{xy} = \frac{2y}{x}$$

$$\partial_{yy} = 2 \ln x$$

(ii)

$$\partial_x = (2x - y)^{-1/2}$$

$$\partial_y = -\frac{1}{2}(2x - y)^{-1/2}$$

$$\partial_{xx} = -(2x - y)^{-3/2}$$

$$\partial_{xy} = \frac{1}{2}(2x - y)^{-3/2}$$

$$\partial_{yy} = -\frac{1}{4}(2x - y)^{-3/2}$$

(iii)

$$\partial_x = (1 - 2x - 8y)e^{-2x} + e^{-3y}$$

$$\partial_y = 4e^{-2x} + (4 - 3x - 12y)e^{-3y}$$

$$\partial_{xx} = 4(-1 + x + 4y)e^{-2x}$$

$$\partial_{xy} = -8e^{-2x} - 3e^{-3y}$$

$$\partial_{yy} = (9x + 36y - 24)e^{-3y}$$